25901201 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR EXAMINATION March 2026

Fourth SEMESTER

Discipline Specific Core Course (DSC) - MG4DSCCFS200 - DATA STRUCTURE USING C++

(2024 ADMISSION ONWARDS)

Duration: 1 hour 30 minutes **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Short Answer Type Questions

Answer any **10** questions

Each question carries **1** marks

1. Describe the "Traversing" operation as it applies to data structures. [U] / [CO1]
2. State the meaning of subscript. [K] / [CO1]
3. Differentiate between insertion at the beginning and at the end of a linked list. [An] / [CO2]
4. What is a linkedlist? [U] / [CO1]
5. In an input-restricted queue, from which end can insertion occur? [An] / [CO2]
6. Examine the relationship between parentheses and stack usage in arithmetic expressions. [U] / [CO2]
7. Identify the principle followed by a stack. [U] / [CO1]
8. Compare the roles of push and pop operations in maintaining stack integrity. [An] / [CO2]
9. Name the variant of a linked list in which none of the node contains NULL pointer. [U] / [CO2]

10. List the data structures, other than arrays, where linear search can be effectively applied. [U] / [CO2]

11. Outline the steps involved in preorder traversal of a binary tree. [An] / [CO2]

12. Predict the index of left child of a node at index 4. [An] / [CO2]

[1x10 = 10]

Part B

Short Essay Type Questions

Answer any **4** questions

Each question carries **5** marks

13. Name and briefly describe three common operations performed on linear data structures. [K] / [CO1]

14. Explain priority queue with memory representation. [U] / [CO1]

15. Give a sequence of stack operations analyze and show the status of the stack after each step. [An] / [CO2]

16. Explain the rule for pushing an operator onto the stack when the stack already contains an operator of equal or higher precedence. [K] / [CO2]

17. Differentiate between strict binary tree and height balanced binary tree. [An] / [CO2]

18. Differentiate between complete binary tree and strict binary tree. [An] / [CO2]

[5x4 = 20]

Part C

Essay Type Questions

Answer any **2** questions

Each question carries **10** marks

19. Describe the representation of a two-dimensional array in memory. Explain how elements are stored linearly even though they are logically arranged in rows and columns. [U] / [CO1]

20. Describe the various steps to evaluate a postfix expression with an example. [U] / [CO2]

21. Explain different types of trees with examples. [U] / [CO2]

[10x2 = 20]